Core Methods in Statistical Computing

23 topics implemented in Python: theory, worked examples and results

BIT 4131 Statistical Computing and Reporting |
Group Members
1. OMBASA ISAAC BSCCS/2024/53672
2. KENNEDY NJUGUNA BSCCS/2024/56648
3. BLESSLEY JOSEPH BSCCS/2023/63927
4. DENNIS MBOGO BSCCS/2024/51424

github.com/Isaac-Y67/statistical-computing

What we did

23 topics	Every topic in the brief, in six batches, from descriptive statistics to reproducible reporting.
Python only	pandas, numpy, scipy, statsmodels, scikit-learn, matplotlib, plotly, Jupyter.
Synthetic data, fixed seeds	True values are known, so each method can be checked against the truth, and every run repeats exactly.
One GitHub repository	A folder per topic: code, data, output tables and plots, and a README with the interpretation.

1 Central tendency and dispersion

THE IDEA

Summarise where data sit (mean, median, mode) and how spread out they are (SD, IQR).

KEY FORMULA

$$s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$$

WHAT WE FOUND

- Mean ≈ median in every subject, so the scores are roughly symmetric.
- Science is the most variable (CV 32.1%); English the most consistent (CV 15.0%).
- The 1.5 × IQR rule flagged the outliers we planted in maths.

Code and results: topic_01_central_tendency_dispersion

2 Probability and conditional probability

THE IDEA

Conditional probability updates a chance once we know B has happened; Bayes reverses the condition.

KEY FORMULA

$$P(A \mid B) = P(A \cap B) / P(B)$$

WHAT WE FOUND

- P(churn | support call) = 0.394 against 0.110 without a call, so they are not independent.
- Bayes' theorem gave P(call | churn) = 0.586, matching the direct count.
- Simulation converged to the data value as draws increased.

Code and results: topic_02_probability

3 Probability distributions

THE IDEA

Model data with Normal, Binomial, Poisson and Exponential distributions, fitted by maximum likelihood.

KEY FORMULA

$$P(X = k) = e^{-\lambda} \lambda^{k} / k!$$

WHAT WE FOUND

- Height fits Normal(169.9, 7.67): K-S p = 0.70, Q-Q plot straight.
- Hand calculations matched Python: P(H < 165) = 0.262, P(no arrivals) = 0.0183.
- AIC slightly favoured gamma for waiting times, but the gap to exponential was small.

Code and results: topic_03_probability_distributions

4 Sampling distributions and the CLT

THE IDEA

Sample means vary from sample to sample; their spread is the standard error, and they become normal as n grows.

KEY FORMULA

$$SE = \sigma / \sqrt{n}$$

WHAT WE FOUND

- Population skewness 2.0; the skewness of sample means fell 0.91 → 0.36 → 0.22 (n = 5, 30, 100).
- Simulated SE 1.829 against theory 1.835 at n = 30.
- Normal approximation gave P(mean > 12) = 0.142; simulation gave 0.141.

Code and results: topic_04_sampling_clt

5 Confidence intervals

THE IDEA

An interval gives plausible values for a parameter; 95% describes the method, not one interval.

KEY FORMULA

$$\bar{x} \pm t^* \cdot s / \sqrt{n}$$

WHAT WE FOUND

- Courier A mean delivery time: 32.23 min, 95% CI (30.65, 33.82).
- Higher confidence gives wider intervals (width 2.64, 3.17, 4.24 for 90, 95, 99%).
- Coverage simulation: the 95% interval captured the true mean 95.0% of the time.

Code and results: topic_05_confidence_intervals

6 Hypothesis testing

THE IDEA

Weigh evidence against H_0 with a test statistic and p-value; report effect size and power too.

KEY FORMULA

$$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$$

WHAT WE FOUND

- Bottle fill: t = −3.248, p = 0.0026, so the mean is below 500 ml.
- Simulated Type I error 4.94% (nominal 5%).
- With 30 per group, power for d = 0.5 is only 0.48; 64 per group is needed for 80%.

Code and results: topic_06_hypothesis_testing

7 Correlation and covariance

THE IDEA

Covariance shows direction; correlation scales it to the range −1 to 1.

KEY FORMULA

$$r = \mathrm{cov}(X, Y) \,/\, (s_x \cdot s_y)$$

WHAT WE FOUND

- Study hours and exam score: r = 0.803, so 64.5% of variation is shared.
- Sleep hours: r = 0.04, no linear relation.
- A U-shaped curve gave r ≈ 0, so always plot the data.

Code and results: topic_07_correlation_covariance

8 Linear regression and least squares

THE IDEA

Fit the line that minimises squared residuals; check the assumptions on the residuals.

KEY FORMULA

$$\hat{\beta} = (X'X)^{-1} X'y$$

WHAT WE FOUND

- Each extra study hour adds 6.08 points; $R^2 = 0.695$.
- Residuals pass normality, constant-variance and autocorrelation checks.
- Test-set $R^2 = 0.668$, close to training, so little overfitting.

Code and results: topic_08_linear_regression

9 Matrix algebra in statistics

THE IDEA

Most statistics are matrix operations: covariance, least squares, leverage, distances.

KEY FORMULA

$$(X'X)\,\hat{\beta} = X'y$$

WHAT WE FOUND

- The normal equations reproduced the regression coefficients exactly.
- The hat matrix is symmetric and idempotent, with trace 5 = number of parameters.
- A singular matrix has no inverse: perfect multicollinearity.

Code and results: topic_09_matrix_algebra

10 Eigenvalues, eigenvectors and PCA

THE IDEA

Eigenvectors of the correlation matrix give uncorrelated axes that capture the most variance.

KEY FORMULA

$$A\,v = \lambda\,v$$

WHAT WE FOUND

- Two components explain 71.6% of variance: quantitative and verbal ability.
- Eigenvalues 2.70 and 2.31 exceed 1 (Kaiser rule).
- Without standardising, reading speed dominated PC1: always standardise.

Code and results: topic_10_eigen_pca

11 Numerical optimization

THE IDEA

Find a minimum step by step when no closed form exists.

KEY FORMULA

$$x_{k+1} = x_k - f'(x_k) \,/\, f''(x_k)$$

WHAT WE FOUND

- Newton's method reached x = 2.25 in 5 iterations (error roughly squares each step).
- Gradient descent diverged for step 0.11, above the 2/20 limit.
- 0 of 100 local searches found the Rastrigin global minimum; differential evolution did.

Code and results: topic_11_numerical_optimization

12 Random numbers and Monte Carlo

THE IDEA

Turn uniform random numbers into estimates; error shrinks like 1/√N.

KEY FORMULA

$$\mathrm{SE} = \sigma / \sqrt{\mathrm{N}}$$

WHAT WE FOUND

- π ≈ 3.14197 from 1,000,000 points; the 95% interval contains π.
- Variance cut 25× (antithetic) and 62× (control variate) on the same budget.
- Importance sampling cut variance 214× for the rare event P(Z > 3).

Code and results: topic_12_random_monte_carlo

13 Time-series analysis

THE IDEA

Split a series into trend, seasonality and noise; difference to make it stationary; forecast.

KEY FORMULA

$$y_t = T_t + S_t + R_t$$

WHAT WE FOUND

- Recovered the trend (2.002 per month, true 2.0) and the seasonal pattern.
- Holt-Winters forecast the held-out year best (MAPE 1.50%); SARIMA 1.70%.
- SARIMA's 95% interval covered only 9 of 12 months, so it was a little too narrow.

Code and results: topic_13_time_series

14 Statistical data visualization

THE IDEA

Match the chart to the question and keep the scales honest.

KEY FORMULA

$$\text{Lie factor} = \text{effect shown} \,/\, \text{effect in data}$$

WHAT WE FOUND

- Log scale showed +4.0% salary per year of experience.
- Truncated bar axis made a true 1.61 ratio look like 3.03.
- Boxplot, violin, ECDF and Q-Q plots each answer a different question.

Code and results: topic_14_statistical_visualization

15 Data transformation and normalization

THE IDEA

Put variables on comparable scales and reduce skew before modelling.

KEY FORMULA

$$z = (x - \bar{x}) / s$$

WHAT WE FOUND

- Three outliers raised the SD by 49.8% but the median by 0%.
- Log cut income skewness from 5.9 to 0.2.
- Only z-scoring gave every variable an equal 25% share of distance.

Code and results: topic_15_data_transformation

16 ANOVA and experimental design

THE IDEA

Test several means at once, then locate differences while controlling the error rate.

KEY FORMULA

$$F = MSB / MSW$$

WHAT WE FOUND

- F = 9.63, p = 3.2×10^{-5}; only Fert B and C differ from control after Tukey.
- Six unadjusted t-tests gave 19.5% false positives; ANOVA stayed at 4.8%.
- Blocking raised the variety F from 0.78 to 8.05.

Code and results: topic_16_anova_experimental_design

17 Chi-square and categorical data

THE IDEA

Compare observed counts with the counts expected under a hypothesis.

KEY FORMULA

$$\chi^2 = \Sigma\,(O - E)^2 \,/\, E$$

WHAT WE FOUND

- Region and preferred product are associated: $\chi^2 = 39.7$, Cramér's V = 0.182.
- Drug against placebo: odds ratio 1.95, risk difference 15.5 points.
- McNemar's test handled paired opinions ($p < 0.0001$).

Code and results: topic_17_chi_square

18 Logistic regression

THE IDEA

Model the log-odds of a yes/no outcome and read coefficients as odds ratios.

KEY FORMULA

$$p = 1 / (1 + e^{(-\eta)})$$

WHAT WE FOUND

- Month-to-month contracts have 2.98 times the odds of churn.
- Test AUC = 0.706: fair ranking, as expected for random outcomes.
- Accuracy 0.667 barely beats always saying 'stayed' (0.663): use recall and AUC.

Code and results: topic_18_logistic_regression

19 Non-parametric methods

THE IDEA

Rank-based tests need fewer assumptions and resist outliers.

KEY FORMULA

$$U_1 = R_1 - n_1(n_1 + 1) / 2$$

WHAT WE FOUND

- Hand-worked U = 21 matched scipy (exact p = 0.0952).
- Mann-Whitney and t-test agreed on skewed data (both p = 0.0019).
- Simulated power on skewed data: 0.847 for Mann-Whitney against 0.454 for the t-test.

Code and results: topic_19_nonparametric

20 Bootstrap and resampling

THE IDEA

Resample the data with replacement to estimate standard errors and intervals.

KEY FORMULA

$$\text{SE_boot} = \text{SD of the } \theta^* \text{ values}$$

WHAT WE FOUND

- Bootstrap SE of the mean 3,514 against the formula 3,500.
- Gave SEs for the median and CV, which have no simple formula.
- On skewed data with n = 10, both intervals covered only 81–84%; the bootstrap fails for the maximum.

Code and results: topic_20_bootstrap_resampling

21 Multivariate analysis

THE IDEA

Analyse correlated variables together: distances, group tests, discrimination, clustering.

KEY FORMULA

$$\Lambda = |W| \,/\, |T|$$

WHAT WE FOUND

- MANOVA: Wilks' Λ = 0.456, p < 0.0001, so species means differ.
- Yet LDA accuracy was 61.7% and clustering recovered little (ARI ≈ 0.1).
- Significant does not mean separable.

Code and results: topic_21_multivariate

22 Missing-data analysis and imputation

THE IDEA

Identify the mechanism (MCAR, MAR, MNAR), then impute and pool with Rubin's rules.

KEY FORMULA

$$T = W + (1 + 1/m)\, B$$

WHAT WE FOUND

- Mean imputation shrank the SD of income from 13.08 to 11.54.
- Under MNAR every method overestimated mean health by 1.4–2.1 points.
- Multiple imputation gives honest SEs under MAR, not under MNAR.

Code and results: topic_22_missing_data

23 Statistical reporting and reproducibility

THE IDEA

Generate the whole report from code so every number can be regenerated.

KEY FORMULA

$$\text{Holm: } p_{adj}(i) = \max_{j \le i} \min(1, (m - j + 1) \cdot p(j))$$

WHAT WE FOUND

- Drug B lowered SBP by 9.6 mmHg against control (95% CI −12.6 to −6.7).
- Dropout differed between arms (p = 0.0067): the main caveat.
- Regenerating from seed 2024 gave an identical data hash.

Code and results: topic_23_reproducible_reporting

What we learned

Check assumptions

Normality, equal variance, independence and sample size decide which method is valid.

Significant is not the same as large or useful

MANOVA was highly significant, yet classification accuracy was only 61.7%.

Match the method to the data

Rank tests, bootstrap and robust scaling won on skewed data; classical tests won on normal data.

Simulation tests our own methods

Coverage, Type I error and power were checked by simulation, and some results surprised us.

Reproducibility

Fixed seeds, one-command rebuilds and a data hash mean anyone can regenerate every number.

Thank you

Questions? All code, data and results are on GitHub:

github.com/Isaac-Y67/statistical-computing